

Mouloud Mammeri University

Department of Sciences

Module: Foreign Language 1

Session 1: Mathematical Vocabulary and Present Simple Tense

Session Objective

The main goal of this session is to master the use of the **Simple Present Tense** to express facts, general truths, and habits, while introducing basic mathematical vocabulary essential for computer science.

The Simple Present Tense (Grammar Review)

The Simple Present is one of the most frequently used tenses in English. It is used to express:

- **Habitual actions:** things we do regularly.
Example: I study computer science every day.
- **General facts or scientific truths.**
Example: Water boils at 100°C.
- **Permanent situations.**
Example: The Earth revolves around the Sun.

Sentence Structure in the Simple Present

Sentence Type	Structure	Example
Affirmative	Subject + Verb (base form)	The Moon reflects the light of the Sun.
Negative	Subject + do/does + not + Verb (base form)	They don't study algebra on holidays.
Interrogative	Do/Does + Subject + Verb (base form)?	Does the Earth revolve around the Sun?

The Third Person Singular (He, She, It)

For the third person singular (*he, she, it*), we add **-s** to the base form of the verb. For all other subjects (*I, you, we, they*), the verb **remains in its base form**.

Examples:

- He plays football.
- She works in a company.
- I play football every weekend.
- They work in a company.

Spelling rules:

- Verbs ending in **s, ss, sh, ch, x, o** → add **-es**: *wash* → *washes*, *go* → *goes*.
- Verbs ending in **consonant + y** → change *y* to *i* and add **-es**: *cry* → *cries*.
- Verbs ending in **vowel + y** → add **-s**: *play* → *plays*.

Essential Auxiliary Verbs

Auxiliary Verb	I	He/She/It	You/We/They
To Be	am	is	are
To Have	have	has	have
To Do	do	does	do

Adverbials of Time for Present Tense

Frequency Adverbs: always, often, sometimes, rarely, never, every day/week, etc.

Time Expressions Indicating Now: now, at the moment, today, these days.

Recurrent Time Phrases: every morning/afternoon/evening, on Mondays, during weekends.

Examples:

- I always check my emails in the morning.
- She never uses social media.

Rules for Adverbials of Time in the Present Simple

- **Position:** Frequency adverbs are usually placed *before the main verb* but *after the verb “to be.”*
 - I **always** study in the evening.
 - She **is never** late.
- **Order of Adverbs:** Time expressions usually come at the *end of the sentence*.
 - They play football **every Sunday**.
- **Use:** These adverbs express **how often** or **when** something happens regularly.

The auxiliary verbs **to be**, **to have**, and **to do** are important in English grammar and are used in the present tense.

Types of Sentences by Purpose and Signs of Punctuation

Types of Sentences by Purpose

Type	Function	Structure
Declarative	States a fact or opinion.	Subject + Verb + Object.
Interrogative	Asks a question.	(Do/Does) + Subject + Verb + Object ?
Imperative	Gives a command or instruction.	Verb (base form) + Object.
Exclamatory	Expresses strong emotion.	What/How + Subject + Verb !

Examples:

- Declarative: The sky is blue.
- Interrogative: Do you like mathematics?
- Imperative: Close the door.
- Exclamatory: What a beautiful day!

Signs of Punctuation

- Period (.) – Ends a statement.
- Question Mark (?) – Ends a question.
- Exclamation Mark (!) – Shows strong emotion.
- Comma (,) – Separates items or clauses.
- Semicolon (;) – Links related clauses or lists.
- Colon (:) – Introduces lists or explanations.
- Apostrophe (') – Shows possession or contractions.
- Quotation Marks (" ") – Encloses direct speech or quotes.
- Parentheses (()) – Adds extra information.
- Hyphen (-) – Joins words or parts.
- Dash (– or —) – Indicates breaks or changes.
- Ellipsis (...) – Shows omission or unfinished thoughts.

General Mathematical Vocabulary

Below is a list of common mathematical symbols used in geometry and algebra, along with their meanings:

- $\equiv$: "is equivalent to" or "is congruent to"
- $\angle$: "measured angle"
- $\perp$: "perpendicular to"
- $\parallel$: "parallel to"
- $\triangle$: "triangle"
- $\cong$: "is congruent to"
- $\sim$: "is similar to"
- π : "pi" (approximately 3.14159)
- $^\circ$: "degree"
- $\odot$: "circle"
- $\emptyset$: "empty set" (no points or elements)

- $|AB|$: "length of line segment AB"
- $\overleftrightarrow{AB}$: "line through two points"
- $\overrightarrow{AB}$: "ray"
- $\cup$: "union of sets" (used in geometry for regions)
- $\cap$: "intersection of sets" (used in geometry for overlapping regions)
- $\subset$: "subset of" (used in set-based geometry)
- R : "radius" of a circle
- d : "diameter" of a circle
- $\neq$: "is not equal to"
- $=$: "is equal to"
- $\rightarrow$: "implies" or "leads to"
- $<$: "is less than"
- $>$: "is greater than"
- $\leq$: "is less than or equal to"
- $\geq$: "is greater than or equal to"
- ∞ : "infinity"
- $+$: "plus" or "added to"
- $\times$: "times" or "multiplied by"
- $\div$: "divided by"